

Sibley School of Mechanical and Aerospace Engineering

MAE 3240 Heat Transfer

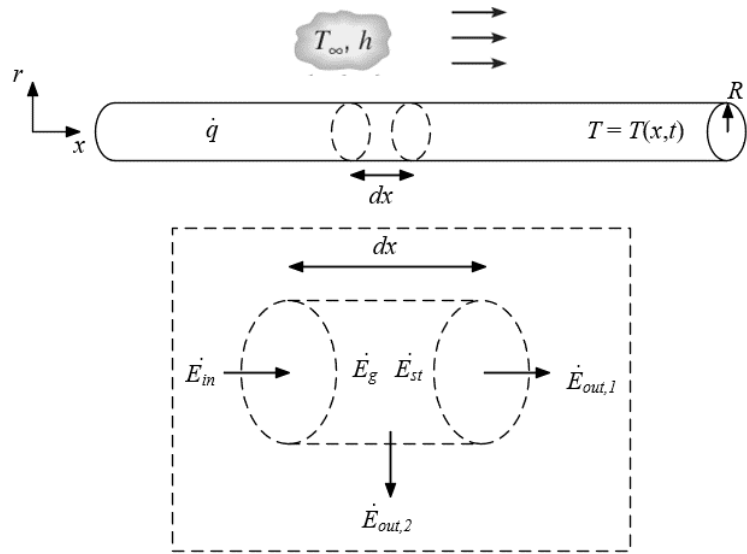
Spring 2026

Problem Set #3

Posted: Monday, Feb 9

Due: Tuesday, Feb 24, 11:59 pm

1. (30 pt) Consider a very long rod with a radius R and a uniform volumetric heat generation rate of $\dot{q}$. This rod is placed in an environment with ambient temperature T_∞ and heat transfer coefficient h . Thermal radiation can be neglected. We set a coordinate system with x -axis through the axis of the rod, as shown in the figure. Assume

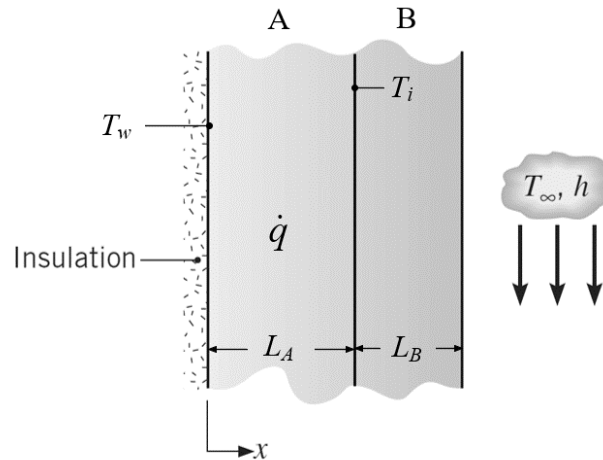


that the radius of the rod is very small and the temperature of the rod along the radial direction is uniform. As a result, the temperature of the rod is a function of the x coordinate and time t , i.e., $T = T(x, t)$. In addition, we can assume the thermal conductivity k , density ρ , and heat capacity c_p are constant. Please revisit the derivation of heat equation in class and answer the following two questions.

- (a) Consider a differential cylinder with a height of dx as the control volume. The figure above shows an energy balance diagram within the control volume. Please write down the energy balance equation for this control volume. Please then express each energy term in the energy balance equation as a function of T by choosing proper rate equations.
- (b) Derive the governing differential equation that describes the temperature profile of the rod and simplify it as much as possible.
- (c) This rod has a length of L . One end of the rod is located at $x = 0$ and maintained at a constant temperature of T_1 . The other end of the rod is located at $x = L$ and exposed to the air flow

with temperature of T_∞ and heat transfer coefficient h . At $t = 0$ s, the rod has a uniform initial temperature of T_i . Please write proper boundary conditions and initial conditions for the governing differential equation.

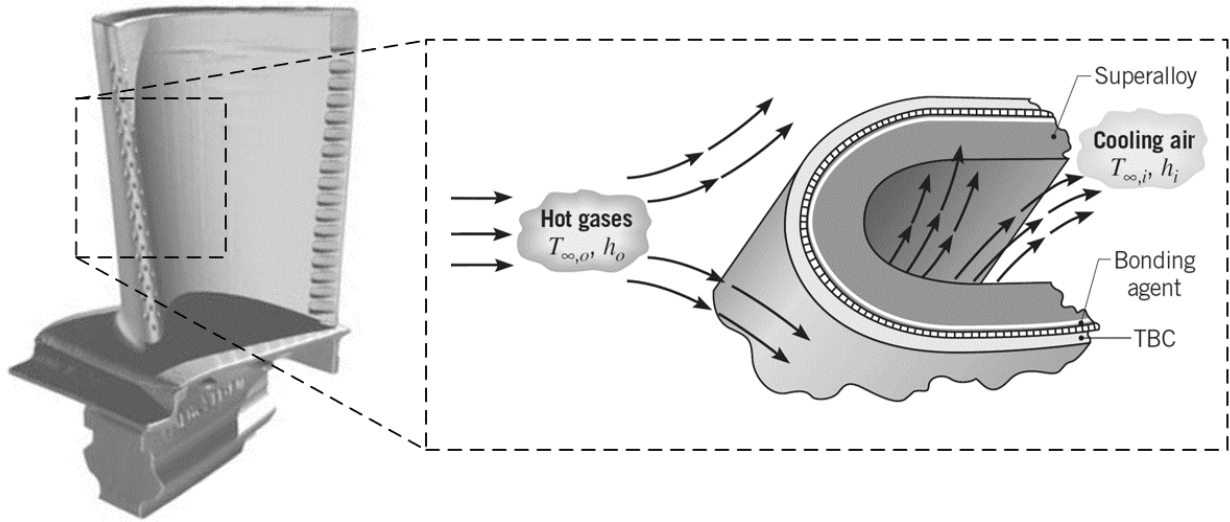
2. (40 pt) Consider one-dimensional heat conduction in a plane composite wall. Wall A has a thickness of $L_A = 30$ cm, which experiences uniform volumetric heat generation $\dot{q}$. The thermal conductivity of wall A is k_A . The left surface of wall A is thermally insulated. Wall B has a thickness of $L_B = 10$ cm and thermal conductivity $k_B = 30$ W/(m·K).



There is no heat generation within wall B. The left surface of wall B is in contact with wall A. There is no contact thermal resistance at their boundary. The right surface of wall B is exposed to a coolant at $T_\infty = 20$ °C and a convective heat transfer coefficient of 500 W/(m²·K). We insert two temperature sensors to measure the surface temperatures of wall A. The sensor at the interface between the thermal insulation and wall A shows $T_w = 115$ °C. The other sensor at the interface between wall A and wall B shows $T_i = 100$ °C.

- Calculate the right surface temperature T_s of wall B.
 - Calculate the uniform volumetric heat generation $\dot{q}$ at steady state.
 - Calculate the thermal conductivity k_A of wall A.
 - Plot the temperature profile across wall A and wall B and show its important features.
3. (30 pt) Energy efficiency of a gas turbine can be improved by operating it under higher temperature. However, hot gases produced by the combustor pose a significant challenge to the reliable operation of turbine blades. An approach to protecting the turbine blades from material failure in the high-temperature environment is to apply a “thermal barrier coating (TBC)” to the exterior surface of the blade, as shown in the figure below. The blade is made from a high-temperature superalloy with thermal conductivity of $k_b = 25$ W/(m·K). The TBC

is made of a ceramic material with thermal conductivity of $k_{TBC} = 1 \text{ W/(m}\cdot\text{K)}$. To attach the TBC to the exterior surface of the superalloy, a metallic bonding agent (thickness negligible) is needed, which creates an interfacial thermal resistance (per unit area) of $R'' = 10^{-4} \text{ m}^2\cdot\text{K/W}$. Consider an operating condition for which external hot gases at $T_{\infty,o} = 1616 \text{ K}$ and cooling air inside the blade at $T_{\infty,i} = 400 \text{ K}$. Heat transfer coefficients of hot gases and cooling air are $h_o = 1000 \text{ W/(m}^2\cdot\text{K)}$ and $500 \text{ W/(m}^2\cdot\text{K)}$, respectively. The turbine blade is at a steady-state condition, where thermal radiation can be neglected. The turbine blade can be approximated as a plane wall, so that we can assume one-dimensional heat conduction through the TBC and the superalloy. The thickness of superalloy blade is $L_b = 5 \text{ mm}$. The maximum allowable temperature for the superalloy is $T_{max} = 1200 \text{ K}$. Consider the following two conditions.



- Condition 1: the TBC is not applied. If we do not apply the TBC and directly expose the superalloy blade to hot gases, draw the thermal resistance network for this condition and label all thermal resistances. Calculate the exterior surface temperature of the superalloy blade. Can the superalloy blade be maintained below T_{max} ?
- Condition 2: the TBC is applied. Now we apply $L_{TBC} = 0.5 \text{ mm}$ thick TBC to the superalloy blade. Draw the thermal resistance network for this condition and label all thermal resistances. Calculate the temperature of the superalloy blade again. Can the superalloy blade be maintained below T_{max} this time?

Cover page

name: Charles Valenbrun

Date of Submission: 2/24/26

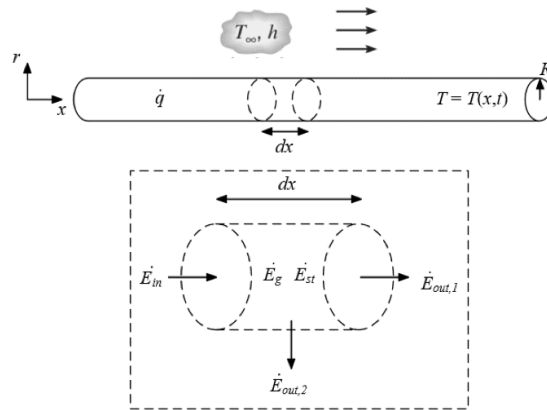
Problem Set # : 3

Due Date: 2/24/26

Acknowledgements :

"I did not give nor receive help"

1. (30 pt) Consider a very long rod with a radius R and a uniform volumetric heat generation rate of $\dot{q}$. This rod is placed in an environment with ambient temperature T_∞ and heat transfer coefficient h . Thermal radiation can be neglected. We set a coordinate system with x -axis through the axis of the rod, as shown in the figure. Assume



that the radius of the rod is very small and the temperature of the rod along the radial direction is uniform. As a result, the temperature of the rod is a function of the x coordinate and time t , i.e., $T = T(x, t)$. In addition, we can assume the thermal conductivity k , density ρ , and heat capacity c_p are constant. Please revisit the derivation of heat equation in class and answer the following two questions.

- Consider a differential cylinder with a height of dx as the control volume. The figure above shows an energy balance diagram within the control volume. Please write down the energy balance equation for this control volume. Please then express each energy term in the energy balance equation as a function of T by choosing proper rate equations.
- Derive the governing differential equation that describes the temperature profile of the rod and simplify it as much as possible.
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with temperature of T_∞ and heat transfer coefficient h . At $t = 0$ s, the rod has a uniform initial temperature of T_i . Please write proper boundary conditions and initial conditions for the governing differential equation.

Known: heat generation $\dot{q}$, radius R , ambient T_∞ , h , k , ρ , c_p

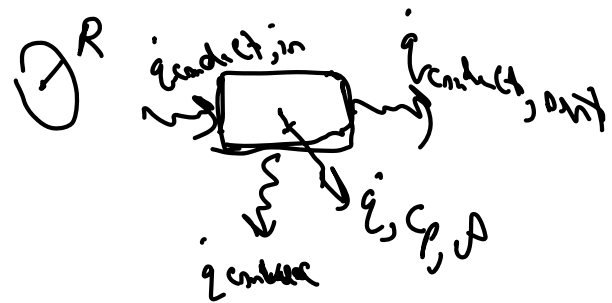
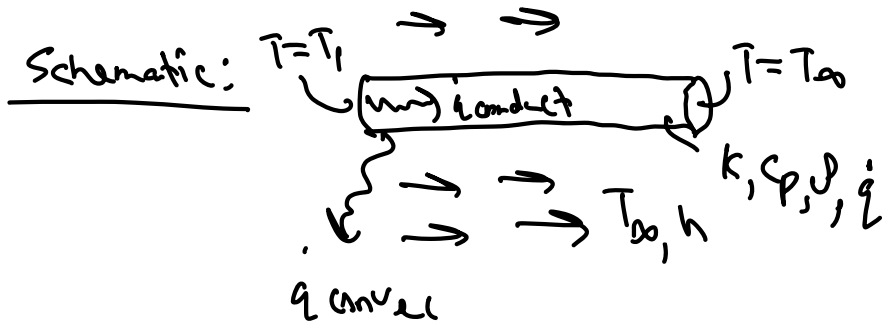
(a) $h, dx \rightarrow$ dimensions of differential volume

(c) rod length L , initial conditions (t) and temperature at each e.d (B.C.S)

Find: (a) energy balance eqn as a function of T

(b) differential eqn of temperature profile

(c) boundary conditions



Assumptions: (1) thermal radiation neglected, (2) const. thermal and physical properties (k, c_p, h, ρ), (3) no work done on system, (4) Temperature doesn't vary radially in rod, (5) $R \ll L$ (very small)

Analysis:

(a) start with energy balance eqn for C.V. $\rightarrow \dot{E}_{in} + \dot{E}_g - \dot{E}_{out} = \dot{E}_{st}$

$$\dot{E}_{in} = \dot{q}_{conduct,in}, \quad \dot{E}_{out} = \dot{q}_{convect} + \dot{q}_{conduct,out}, \quad \dot{E}_g = \dot{q}V = \dot{q}\pi R^2 dx$$

$$\dot{E}_{st} = \int \rho c_p \frac{\partial T}{\partial t} \pi R^2 dx$$

plugging into energy eqn: $\dot{q}_{conduct,in} - \dot{q}_{convect} - \dot{q}_{conduct,out} + \dot{q}\pi R^2 dx = \int \rho c_p \frac{\partial T}{\partial t} \pi R^2 dx$

For heat rates: $\dot{q}_{conduct,in} = -kA \frac{\partial T}{\partial x} = -k\pi R^2 \frac{\partial T}{\partial x}$

$$\dot{q}_{convect} = hA(T - T_{\infty}) = h(2\pi R dx)(T - T_{\infty})$$

$$\dot{q}_{conduct,out} = -k(\pi R^2) \left(\frac{\partial T}{\partial x} + \frac{\partial^2 T}{\partial x^2} dx \right)$$

(b) (continuing from (a)), we can plug in all expressions into energy balance

$$\begin{aligned} \hookrightarrow -k\pi R^2 \frac{\partial T}{\partial x} - k(\pi R^2) \left(\frac{\partial T}{\partial x} + \frac{\partial^2 T}{\partial x^2} dx \right) - h(2\pi R dx)(T - T_{\infty}) + \dot{q}\pi R^2 dx \\ = \int \rho c_p \frac{\partial T}{\partial t} \pi R^2 dx \end{aligned}$$

Combining Conduction terms (leaves):

$$k\pi R^2 dx \frac{\partial^2 T}{\partial x^2} + 2\pi R h dx (T_\infty - T) + \dot{q}\pi R^2 dx = \rho c_p \frac{\partial T}{\partial t} \pi R^2 dx$$

$$\hookrightarrow \boxed{k \frac{\partial^2 T}{\partial x^2} + \dot{q} + \frac{2}{R} h (T_\infty - T) = \rho c_p \frac{\partial T}{\partial t}}$$

(C) T is a function of both time and position along rod

$$\hookrightarrow f = T(x, t)$$

Boundary conditions: at $x=0$, $T=T_i$ regardless of time

$$\text{that is } \boxed{T(x=0, t) = T_i}$$

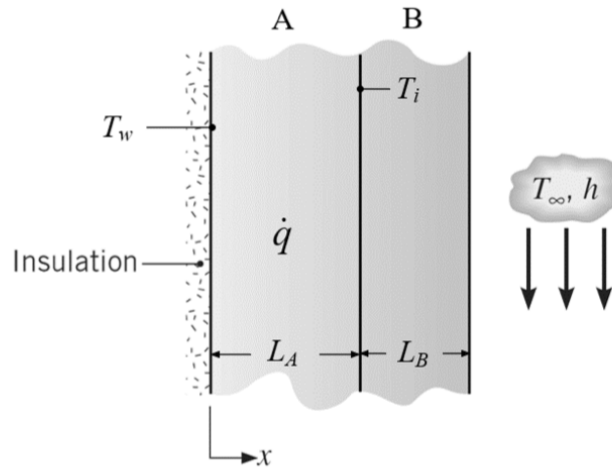
at $x=L$, Conduction = Convection $\rightarrow -k \frac{\partial T}{\partial x} = h[T(L, t) - T_\infty]$

$$\hookrightarrow \boxed{T(L, t) = \frac{-k}{h} \frac{\partial T}{\partial x} + T_\infty}$$

Initial conditions: at $t=0$, $T=T_i$ that's uniform throughout

$$\text{the rod} \rightarrow \boxed{T(x, t=0) = T_i}$$

2. (40 pt) Consider one-dimensional heat conduction in a plane composite wall. Wall A has a thickness of $L_A = 30$ cm, which experiences uniform volumetric heat generation $\dot{q}$. The thermal conductivity of wall A is k_A . The left surface of wall A is thermally insulated. Wall B has a thickness of $L_B = 10$ cm and thermal conductivity $k_B = 30$ W/(m·K).



There is no heat generation within wall B. The left surface of wall B is in contact with wall A. There is no contact thermal resistance at their boundary. The right surface of wall B is exposed to a coolant at $T_\infty = 20$ °C and a convective heat transfer coefficient of 500 W/(m²·K). We insert two temperature sensors to measure the surface temperatures of wall A. The sensor at the interface between the thermal insulation and wall A shows $T_w = 115$ °C. The other sensor at the interface between wall A and wall B shows $T_i = 100$ °C.

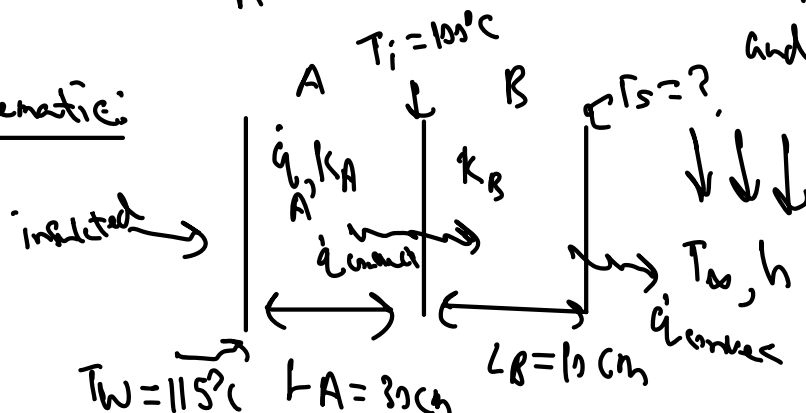
- Calculate the right surface temperature T_s of wall B.
- Calculate the uniform volumetric heat generation $\dot{q}$ at steady state.
- Calculate the thermal conductivity k_A of wall A.
- Plot the temperature profile across wall A and wall B and show its important features.

known : $L_A = 30$ cm, $L_B = 10$ cm, $k_B = 30 \frac{\text{W}}{\text{m}\cdot\text{K}}$, $\dot{q}_B = 0$, $T_\infty = 20^\circ\text{C}$
 $h = 500 \frac{\text{W}}{\text{m}^2\cdot\text{K}}$, $T_w = 115^\circ\text{C}$, $T_i = 100^\circ\text{C}$

Find : (a) T_s of right surface (c) k_A of wall A

(b) $\dot{q}_A$ in steady state (d) temperature profile across wall A and wall B

Schematic:



Assumptions: (1) thermal properties (h, k) are uniform in materials, (2) no work is done on system, (3) neglect PE and KE, (4) heat transfer is 1-D, (5) no contact thermal resistance at boundaries (6) neglect thermal radiation

Analysis:

(a) To find T_s , we can use heat flux analysis at the rightmost surface of Wall B $\rightarrow \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g^D = \dot{E}_f^D$

$$\hookrightarrow \dot{E}_{in} = \dot{E}_{out} \rightarrow q_{in} = q_{out} \rightarrow q''_{\text{conduct}} = q''_{\text{convect}}$$

$$q''_{\text{conduct}} = -k_B \frac{dT}{dx} = -k_B \frac{T_s - T_i}{L_B}$$

$$q''_{\text{convect}} = h(T_s - T_{\infty})$$

Conduction between wall A-B interface

Convection between outer wall and B

$$\hookrightarrow k_B \frac{T_i - T_s}{L_B} = h(T_s - T_{\infty}) \Rightarrow T_s = \frac{\frac{k_B}{L_B} T_i + h T_{\infty}}{\frac{k_B}{L_B} + h}$$

$$T_s = \frac{\frac{30 \frac{\text{W}}{\text{mK}}}{10 \text{ cm}} (10 + 273.15 \text{ K}) + 500 \frac{\text{W}}{\text{m}^2 \text{K}} (20 + 273.15 \text{ K})}{\frac{30 \frac{\text{W}}{\text{mK}}}{10 \text{ cm}} + 500 \frac{\text{W}}{\text{m}^2 \text{K}}} = \boxed{323.15 \text{ K} = 50^\circ \text{C}} = T_s$$

(b) We now look at a C.V. Surrounding wall A to find

$$\dot{q}_A \rightarrow \dot{E}_{in} - \dot{E}_{out} + \dot{E}_g = \dot{E}_f \rightarrow \text{since Steady State}$$

$\rightarrow \text{since leftmost surface is insulated}$

$$\hookrightarrow \dot{E}_{out} = \dot{E}_g \rightarrow \dot{q}_A (A \cdot L_A) = q_{\text{conduct}}$$

$$q_{\text{conduct}} = -k_A \frac{T_s - T_i}{L_A} \rightarrow \dot{q}_A (A \cdot L_A) = k_A \frac{T_i - T_s}{L_B}$$

$$\rightarrow \dot{q}_A = \frac{k_B}{L_A L_B} (T_i - T_s) = \frac{30 \frac{\text{W}}{\text{mK}}}{30 \text{ cm} \cdot 10 \text{ cm}} (100 - 50^\circ \text{C}) = \boxed{50,000 \frac{\text{W}}{\text{mK}}}$$

(c) Temperature profile in wall governed by differential eqn:

$$k_A \frac{d^2 T}{dx^2} + \dot{q} = 0 \text{ with 1-D and steady state}$$

$$\hookrightarrow \frac{dT}{dx} = -\frac{\dot{q}}{k_A} x + C_1, \text{ rightmost wall is insulated} \Rightarrow T'(x=0) = 0$$

$$\text{So } C_1 = 0$$

$$\hookrightarrow T_A(x) = -\frac{\dot{q}}{2k_A} x^2 + C_2, T(x=0) = 115^\circ \text{C} = 388.15 \text{ K}$$

$$388.15 \text{ K} = \frac{-50,000 \frac{\text{W}}{\text{m}^2}}{2k_A} (0) + C_2 \Rightarrow C_2 = 388.15 \text{ K}$$

$$T(x=30 \text{ cm}) = 100^\circ \text{C} = 373.15 \text{ K} = \frac{-50,000}{2k_A} (30 \text{ cm})^2 + 388.15 \text{ K}$$

$$\hookrightarrow k_A = \frac{-50,000 \frac{\text{W}}{\text{m}^2} (30 \text{ cm})^2}{2(373.15 - 388.15 \text{ K})} = \boxed{150 \frac{\text{W}}{\text{mK}}}$$

(d) in wall B, governing differential eqn is $k_B \frac{dT}{dx} = 0$

$$\rightarrow T_B(x) = C_1 x + C_2 \rightarrow T_2(x=30 \text{ cm}) = 373.15 \text{ K} = C_1 (30 \text{ cm}) + C_2$$

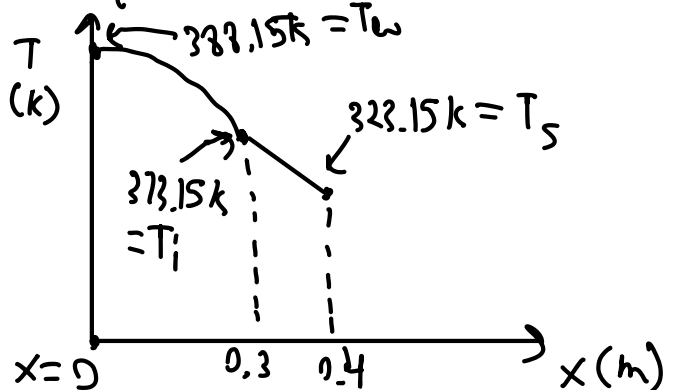
$$\text{at } T_B(x=40 \text{ cm}) = 323.15 \text{ K} = C_1 (40 \text{ cm}) + C_2$$

$$\hookrightarrow 373.15 \text{ K} - 323.15 = C_1 (30 \text{ cm} - 40 \text{ cm})$$

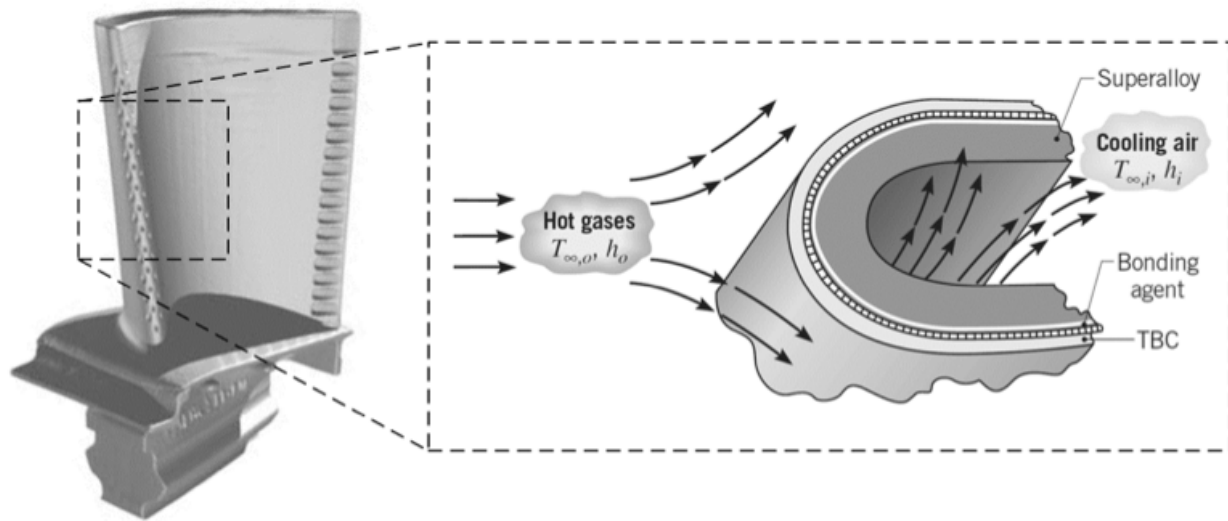
$$\hookrightarrow C_1 = -500 \frac{\text{K}}{\text{m}} \Rightarrow C_2 = 323.15 + 500(40 \text{ cm}) = 523.15 \text{ K}$$

$$T_B(x) = -500 \frac{\text{W}}{\text{mK}} x + 523.15 \text{ K}$$

$$T_A(x) = \frac{-500}{2} x^2 + 388.15 \text{ K}$$



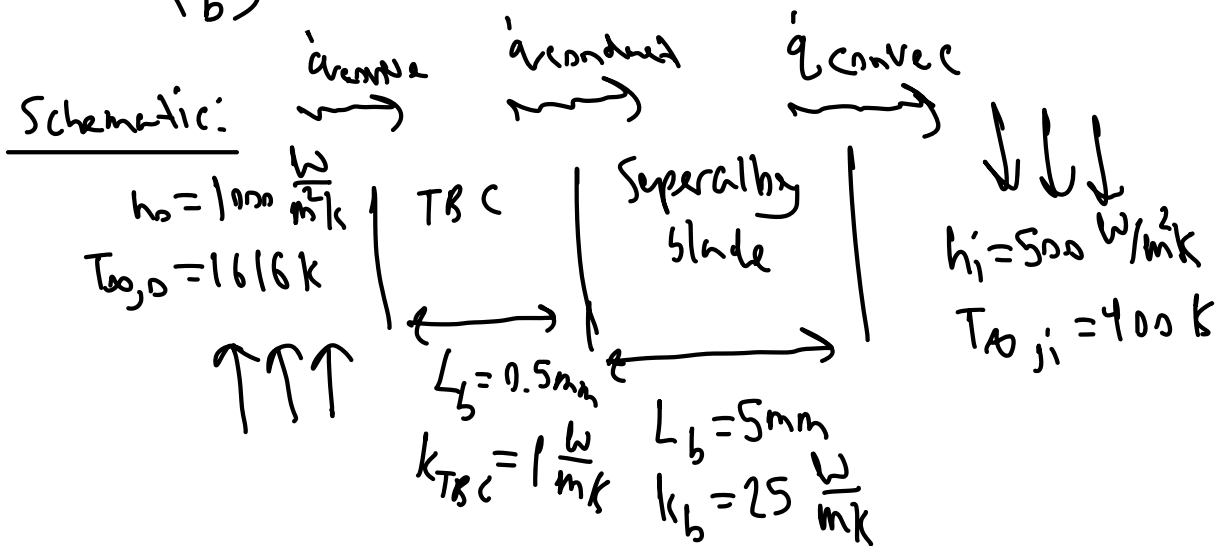
3. (30 pt) Energy efficiency of a gas turbine can be improved by operating it under higher temperature. However, hot gases produced by the combustor pose a significant challenge to the reliable operation of turbine blades. An approach to protecting the turbine blades from material failure in the high-temperature environment is to apply a “thermal barrier coating (TBC)” to the exterior surface of the blade, as shown in the figure below. The blade is made from a high-temperature superalloy with thermal conductivity of $k_b = 25 \text{ W/(m}\cdot\text{K)}$. The TBC is made of a ceramic material with thermal conductivity of $k_{TBC} = 1 \text{ W/(m}\cdot\text{K)}$. To attach the TBC to the exterior surface of the superalloy, a metallic bonding agent (thickness negligible) is needed, which creates an interfacial thermal resistance (per unit area) of $R'' = 10^{-4} \text{ m}^2\cdot\text{K/W}$. Consider an operating condition for which external hot gases at $T_{\infty,o} = 1616 \text{ K}$ and cooling air inside the blade at $T_{\infty,i} = 400 \text{ K}$. Heat transfer coefficients of hot gases and cooling air are $h_o = 1000 \text{ W/(m}^2\cdot\text{K)}$ and $500 \text{ W/(m}^2\cdot\text{K)}$, respectively. The turbine blade is at a steady-state condition, where thermal radiation can be neglected. The turbine blade can be approximated as a plane wall, so that we can assume one-dimensional heat conduction through the TBC and the superalloy. The thickness of superalloy blade is $L_b = 5 \text{ mm}$. The maximum allowable temperature for the superalloy is $T_{max} = 1200 \text{ K}$. Consider the following two conditions.



- (a) Condition 1: the TBC is not applied. If we do not apply the TBC and directly expose the superalloy blade to hot gases, draw the thermal resistance network for this condition and label all thermal resistances. Calculate the exterior surface temperature of the superalloy blade. Can the superalloy blade be maintained below T_{max} ?
- (b) Condition 2: the TBC is applied. Now we apply $L_{TBC} = 0.5 \text{ mm}$ thick TBC to the superalloy blade. Draw the thermal resistance network for this condition and label all thermal resistances. Calculate the temperature of the superalloy blade again. Can the superalloy blade be maintained below T_{max} this time?

known: $k_b = 25 \frac{W}{mK}$, $k_{TBC} = 1 \frac{W}{mK}$, $R'' = 10^{-4} \frac{m^2 K}{W}$, $T_{\infty, o} = 1616 K$
 $T_{\infty, i} = 400 K$, $h_o = 1000 \frac{W}{m^2 K}$, $h_i = 500 \frac{W}{m^2 K}$, $L_b = 5 mm$, $T_{max} = 1200 K$
 (b) $L_{TBC} = 0.5 mm$

Find: $\begin{pmatrix} u \\ k \\ b \end{pmatrix}$ thermal resistance network, exterior surface temperature T_{ext}



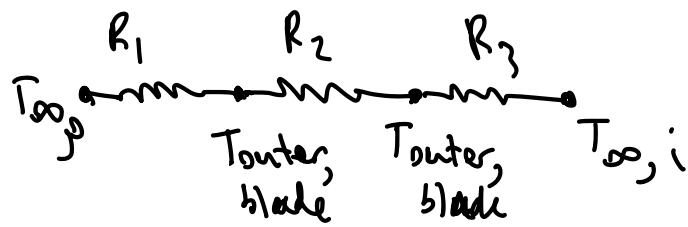
Assumptions: (1) no work done on system, (2) PE and KE neglected,
 (3) thermal radiation negligible, (4) steady state, (5) 1-D thermal flow
 (6) thermal properties const throughout material

Analysis:

(a) without the TBC and bonding agent, flow of heat would go from hightemp. gas $\rightarrow$ Superalloy blade $\rightarrow$ coolant fluid

$$\rightarrow T_{\infty, o} \mid T_{outer, blade} \mid T_{inner, blade} \mid T_{\infty, i}$$

The thermal resistance network would then be



where $R_1 = \frac{1}{h_{\infty,o} A}$, $R_2 = \frac{L_b}{k_b A}$

$R_3 = \frac{1}{h_{\infty,i} A}$

effective thermal resistance is $R_1 + R_2 + R_3 = R_{eff}$

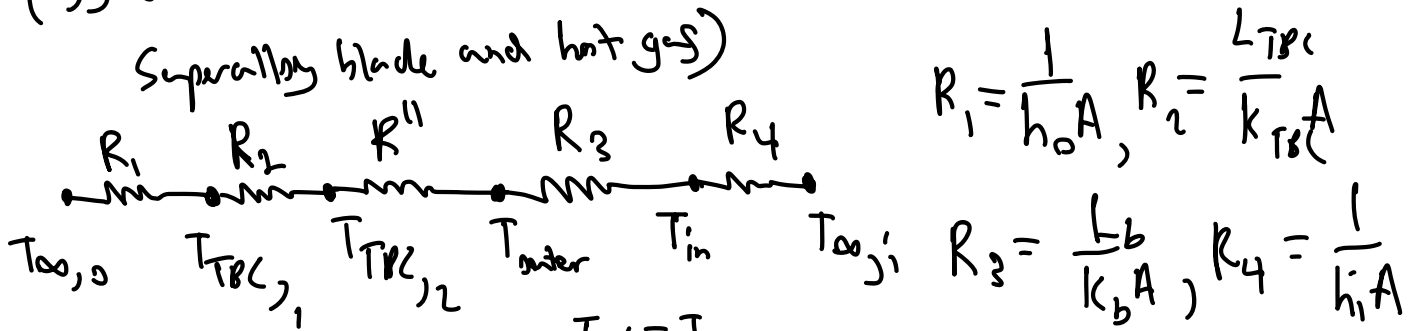
The heat rate through the network is $q = \frac{\Delta T}{R_{th}} = \frac{T_{\infty,i} - T_{\infty,o}}{\frac{1}{A} \left(\frac{1}{h_i} + \frac{L_b}{k_b} + \frac{1}{h_o} \right)}$

$T_{outer} - T_{\infty,o} = q R_1 = \frac{T_{\infty,i} - T_{\infty,o}}{\frac{1}{A} \left(\frac{1}{h_i} + \frac{L_b}{k_b} + \frac{1}{h_o} \right) \frac{1}{h_o A}}$

$T_{outer} = T_{\infty,o} + \frac{T_{\infty,i} - T_{\infty,o}}{h_o \left(\frac{1}{h_i} + \frac{L_b}{k_b} + \frac{1}{h_o} \right)} = 1616 K + \frac{400 - 1616 K}{1000 \frac{W}{m^2 K} \left(\frac{1}{500 \frac{W}{m^2 K}} + \frac{(5mm)}{25 \frac{W}{mK}} + \frac{1}{1000} \right)}$

$T_{outer} = 1236 K \rightarrow$ greater than $T_{max} = 1200 K$ So cannot be maintained lower than T_{max}

(b) we now add TBC and bonding to thermal network (in between Superalloy blade and hot gas)



$R_1 = \frac{1}{h_o A}$, $R_2 = \frac{L_{TBC}}{k_{TBC} A}$

$R_3 = \frac{L_b}{k_b A}$, $R_4 = \frac{1}{h_i A}$

heat rate through network: $q = \frac{T_{\infty,i} - T_{\infty,o}}{R_1 + R_2 + \frac{R''}{A} + R_3 + R_4}$

$\rightarrow T_{outer} = T_{\infty,o} + \frac{T_{\infty,i} - T_{\infty,o}}{R_1 + R_2 + \frac{R''}{A} + R_3 + R_4} = T_{\infty,o} + \frac{T_{\infty,i} - T_{\infty,o}}{\left(\frac{1}{h_o} + \frac{L_{TBC}}{k_{TBC}} + \frac{L_b}{k_b} + \frac{1}{h_i} + R'' \right)}$

$$T_{\text{inter}} = 1616 \text{ K} + \frac{(4000 \text{ K} - 1616 \text{ K}) \left(\frac{1}{1000 \frac{\text{W}}{\text{m}^2 \text{K}}} + \frac{0.5 \text{ mm}}{1 \frac{\text{W}}{\text{mK}}} + 10 \frac{\text{m}^2 \text{K}}{\text{W}} \right)}{\left(\frac{1}{1000 \frac{\text{W}}{\text{m}^2 \text{K}}} + \frac{0.5 \text{ mm}}{1 \frac{\text{W}}{\text{mK}}} + \frac{5 \text{ mm}}{25 \frac{\text{W}}{\text{mK}}} + \frac{1}{500 \frac{\text{W}}{\text{m}^2 \text{K}}} + 10 \frac{\text{m}^2 \text{K}}{\text{W}} \right)}$$

$$\boxed{T_{\text{inter}} = 1104 \text{ K}} \rightarrow T_{\text{inter}} < T_{\text{max}} = 1200 \text{ K}$$

So the Superalloy blades under this condition can be maintained under T_{max}